# Binomial random variables

**1** In a binomial experiment:

**a** Are the trials independent or dependent?

**b** How many possible outcomes are there for each trial?

**2** State whether following scenarios describe a binomial random variable:

**a** Rolling a six-sided die $20$ times and counting the number of times the die lands on $1$.

**b** The sum of the outcomes of rolling a six-sided die $10$ times.

**c** Selecting $70$ random people from the population and recording the number of females.

**d** A particular coin is more likely to land on heads than tails. Tossing this coin $10$ times and recording the number of times it lands on heads.

**e** Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of purple marbles.

**f** Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of red marbles.

**3** Consider the binomial probability formula:

$$P(x) = {}^{n}C_{x}p^{x}(1-p)^{n-x}$$

where $P(x)$ is the probability of obtaining $x$ successes in $n$ independent trials.

**a** What does $p$ represent in the formula?

**b** What does $(1-p)$ represent in the formula?

**4** For a binomial probability distribution with $n$, trials and probability of success $p$, Ned performed the following calculation:

$$\binom{n}{2}p^2 \times 0.7^4$$

**a** State the probability of success.

**b** State the number of trials.

**c** What probability is Ned trying to find with this calculation?

**5** Consider the binomial power $(p+q)^4$.

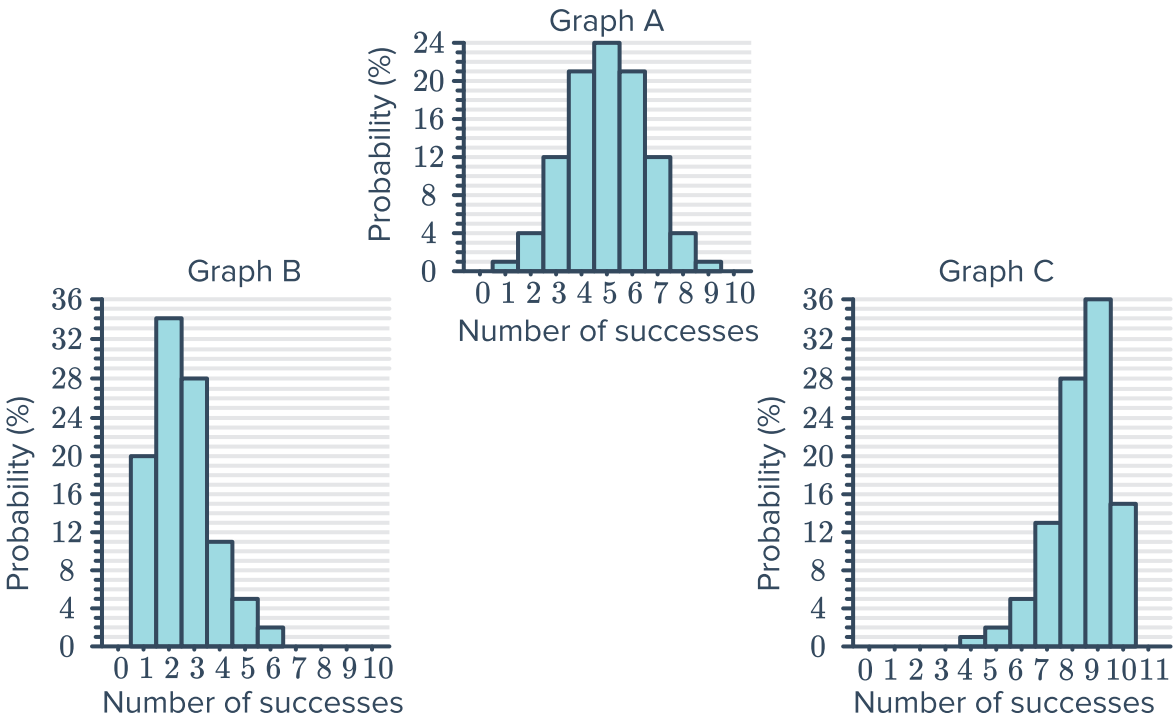
**a** Expand $(p+q)^4$.

**b** We can interpret the terms in the expansion in terms of probabilities in a binomial distribution, with $p$ the probability of success and $q$ the probability of failure. What probability would the term $\binom{4}{2}p^2q^2$ in the expansion represent?

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## Distributions of binomial random variables

**6** Consider the three graphs below that show binomial distributions for $n = 10$ and $p = 0.15$, $0.5$ and $0.85$:




**a** Which value of $p$ corresponds to Graph A?

**b** Which value of $p$ corresponds to Graph B?

**c** Which value of $p$ corresponds to Graph C?

**7** Consider a binomial distribution of a random variable $X$ with $p = 0.5$ and $n = 17$.

**a** Is $P(\mathrm{X} = 5)$ greater than, less than or equal to $P(\mathrm{X} = 12)$?

**b** Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

**8** Consider a binomial distribution of a random variable $X$ with $p = 0.13$ and $n = 11$.

**a** Is $P(\mathrm{X} = 4)$ greater than, less than or equal to $P(\mathrm{X} = 7)$?

**b** Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

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**9** Consider a binomial distribution of a random variable $X$ with $p = 0.83$ and $n = 17$.

**a** Is $P(\mathrm{X} = 2)$ greater than, less than or equal to $P(\mathrm{X} = 15)$?

**b** Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

**10** For each of the followng table of values for the binomial distribution $X$:

**i** Describe the shape of the graph as positively skewed, symmetric, or negatively skewed.

**ii** Calculate the value of $p$ correct to two decimal places.

**a**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.004 | 0.031 | 0.109 | 0.219 | 0.274 | 0.219 | 0.109 | 0.031 | 0.004 |

**b**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.152 | 0.323 | 0.3 | 0.16 | 0.053 | 0.011 | 0.001 | 0 | 0 |

**c**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $P(X = x)$ | 0 | 0 | 0.001 | 0.009 | 0.046 | 0.147 | 0.294 | 0.336 | 0.168 |

## Probability for binomial random variables

**11** Find the value of the following:

**a** ${}^5C_4 \times (0.1)^4 \times 0.9 + {}^5C_5 \times (0.1)^5 \times (0.9)^0$

**b** ${}^5C_3 \times (0.8)^3 \times (0.2)^2 + {}^5C_4 \times (0.8)^4 \times 0.2 + {}^5C_5 \times (0.8)^5 \times (0.2)^0$

**12** A binomial random variable $X$ has 13 independent trials and a probability of success of 0.4.

**a** Using the definition for the binomial random variable $X$, write the formula for $P(X = 2)$.

**b** Calculate $P(X = 2)$ to four decimal places.

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**13** For a binomial random variable $X$, $P(X=3) = {}^{10}C_3 \times 0.6^3 \times 0.4^7$.

**a** Calculate $P(X=3)$ to four decimal places.

**b** State the number of trials in this experiment.

**c** State the probability of success in this experiment.

**14** $X$ is a binomial variable with the probability mass function:

$$P(X=k) = {}^4C_k \times (0.4)^k \times (0.6)^{4-k} \text{ for } k = 0,1,2,3,4$$

**a** State the number of trials for this distribution.

**b** State the probability of success.

**c** Wrie an expression for $P(X=2)$.

**d** How many ways can we get 2 successes in the 4 trials?

**e** Calculate the probability $P(X=2)$.

**15** $X$ is a binomial variable with the probability mass function:

$$P(X=k) = {}^3C_k \times (0.3)^k \times (0.7)^{3-k} \text{ for } k = 0,1,2,3$$

Complete the table:

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X=x)$ | 0.343 | 0.441 | | |

**16** Consider the random variable $X$, which has distribution $X \sim B(4, 0.1)$.

**a** Write an expression for $P(X=1)$.

**b** Write an expression for $P(X=2)$.

**c** Complete the following table:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X=x)$ | 0.6561 | | | | 0.0001 |

**17** A binomial random variable $X$ is defined as:

$$P(X=x) = {}^{20}C_x \times 0.75^x \times 0.25^{20-x} \text{ for } x = 1,2,\ldots,20$$

**a** State the number of trials in this experiment.

**b** State the probability of success in this experiment.

**c** Calculate $P(X=11)$ to four decimal places.

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$$P(X = k) = {}^{6}C_k \times (0.8)^k \times (0.2)^{6-k} \text{ for } k = 0, 1, 2, 3, 4, 5, 6$$

**a** State the number of trials for this distribution.

**b** What is the probability of failure?

**c** Calculate the probability $P(X = 3)$.

**d** Calculate the probability $P(X \leq 3)$.

**19** $X$ is a binomial variable with the probability mass function:

$$P(X = k) = {}^{5}C_k \times (0.4)^k \times (0.6)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

**a** Calculate $P(X \geq 3)$.

**b** Calculate $P(X < 3)$.

**20** $X$ is a binomial variable with the probability mass function:

$$P(X = k) = {}^{5}C_k \times (0.2)^k \times (0.8)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

**a** State the number of trials for this distribution.

**b** State the probability of success.

**c** Calculate the probability $P(X > 3)$.

**d** What is the most likely number of successes?

**e** Calculate the mean of the distribution.

**f** Calculate the variance of the distribution.

**21** Consider the random variable $X$, which has distribution $X \sim B\left(8, \frac{4}{5}\right)$.

**a** State the number of trials for this distribution.

**b** State the probability of success.

**c** Calculate the probability $P(X > 4)$ to two decimal places.

**d** What is the most likely number of successes?

**e** Calculate the mean of the distribution.

**f** Calculate the variance of the distribution.

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**22** $X$ is a binomial variable with a probability distribution given in the table:

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | 0.00243 | 0.02835 | 0.1323 | 0.3087 | 0.36015 | 0.16807 |

**a** Find $P(X \geq 4)$.

**b** Find $P(X < 3)$.

**23** The following table represents a probability distribution of a binomial random variable:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.2401 | 0.4116 | 0.2646 | 0.0756 | 0.0081 |

**a** State the number of trials for this distribution.

**b** Calculate the probability $P(X = 3)$.

**c** Calculate the probability $P(X < 3)$.

**d** Using $P(X = 4)$, calculate the probability of success $p$.

**e** Calculate $E(X)$.

**f** Calculate the standard deviation of the distribution.

**24** The table represents a cumulative probability distribution of a binomial random variable:

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(X \leq x)$ | 0.01024 | 0.08704 | 0.31744 | 0.66304 | 0.92224 | 1 |

**a** State the number of trials for this distribution.

**b** Calculate $P(X \leq 2)$.

**c** Calculate $P(X \geq 3)$.

**d** Calculate $P(X = 5)$.

**e** Using $P(X = 5)$, calculate the probability of success $p$.

**f** Calculate the standard deviation of the distribution.

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**25** The following graph represents a probability distribution of a binomial random variable:

**a** State the number of trials for this distribution.

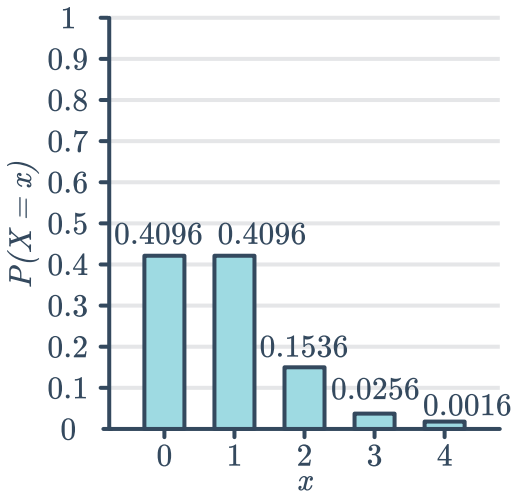
**b** Calculate $P(X = 3)$.

**c** Calculate $P(X < 3)$.

**d** Using $P(X = 4)$, calculate the probability of success $p$.

**e** Calculate $E(X)$.

**f** Calculate the standard deviation of the distribution.




**26** The following graph represents a cumulative probability distribution of a binomial random variable:

**a** State the number of trials for this distribution.

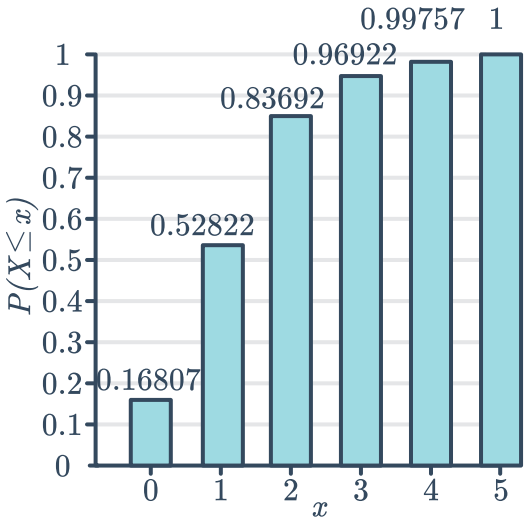
**b** Calculate $P(X \leq 2)$.

**c** Calculate $P(X \geq 3)$.

**d** Calculate $P(X = 5)$.

**e** Using $P(X = 5)$, calculate the probability of success $p$.

**f** Calculate the standard deviation of the distribution.




**27** Calculate the probability of the following, rounding your answers to two decimal places:

**a** Getting exactly $18$ successes in $30$ independent trials if the probability of success in each trial is $0.7$.

**b** Getting at least $10$ successes in $80$ independent trials if the probability of success in each trial is $p = 0.1$.

**c** Getting at most $22$ successes in $25$ independent trials if the probability of success in each trial is $0.85$.

**d** Getting more than $15$ successes in $33$ independent trials if the probability of success in each trial is $0.45$.

**e** Getting less than $12$ successes in $50$ independent trials if the probability of success in each trial is $0.28$.

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**28** $P(X = 2) = 0.3456$ for a binomial random variable $X$ with $4$ trials.

**a** If success is more likely, find the probability of success $p$.

**b** Calculate $E(X)$.

**c** Calculate $\text{Var}(X)$.

**29** $E(X) = 6$ for a binomial random variable $X$ with $10$ trials.

**a** Find the probability of success $p$.

**b** Calculate $\text{Var}(X)$.

**30** The expected value of a binomial distribution is $8$ and the variance is $6.4$.

**a** Find the number of trials, $n$.

**b** Find the probability of success, $p$.

**31** $E(X) = \dfrac{15}{2}$ for a binomial random variable $X$ with probability of success $\dfrac{3}{10}$.

**a** Find the number of trials $n$.

**b** Calculate the standard deviation of $X$.

**32** $E(X) = 2$ and $\text{Var}(X) = 1.6$ for a binomial random variable $X$.

**a** Find the probability of success $p$.

**b** Find the number of trials $n$.

**33** $E(X) = 4.8$ and $\sigma(X) = \sqrt{2.88}$ for a binomial random variable $X$.

**a** Find the probability of success $p$.

**b** Find the number of trials $n$.

## Applications

**34** A certain disease has a survival rate of $64\%$. Of the next $110$ people who contract the disease, how many would you expect to survive? Round your answer to the nearest whole number.

**35** The probability of a particular tennis player getting their first serve in is $0.72$. If she serves $90$ times in a match, how many times would she expect to get her first serve in? Round your answer rounded to the nearest whole number.

**36** A fair die is rolled $5$ times and the number of fours noted.

**a** On each trial, what is the probability of success?

**b** Calculate the probability of rolling exactly $2$ fours, correct to four decimal places.

**c** Calculate the probability of rolling exactly $3$ fours, correct to four decimal places.

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**37** David owns $3$ yellow mugs and $8$ blue mugs. Each morning, he chooses a mug at random for his cup of tea before work, $5$ days a week. The mug is washed and returned before the next day.

**a** Each day, what is the probability of choosing a yellow mug?

**b** What is the probability that he used a yellow mug exactly twice this past work week? Round your answer to four decimal places.

**c** What is the probability that he never used a yellow mug this past work week? Round your answer to four decimal places.

**38** At a local Italian restaurant, $16$ people are in line to order. The probability that any one person will choose pizza is $23\%$.

**a** What is the probability that exactly $4$ people choose pizza? Round your answer to four decimal places.

**b** What is the probability that exactly half of the people choose pizza? Round your answer to four decimal places.

**39** Consider the binomial $\left(\frac{1}{5}+\frac{4}{5}\right)^5$.

**a** Write the formula for the term of the expansion that contains $\left(\frac{4}{5}\right)^3$.

**b** A box of chocolates is $\frac{1}{5}$ dark chocolate and $\frac{4}{5}$ milk chocolate. If $5$ chocolates are selected at random with replacement, describe what the term from part (a) would tell us.

**c** Calculate the probability of selecting exactly $1$ dark chocolate.

**d** Calculate the probability of selecting at most $1$ dark chocolate.

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